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Lightweight Scene Understanding and Synthetic Data-Driven Damage Inspection for Martian Rovers

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for Martian Rovers**

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Dedicato a
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Abstract

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Chapter 1

Synthetic Dataset Generation

1.1 Design Requirement

The dataset was designed to contain RGB views covering all six wheel positions of a Curiosity Rover model, with one designated target wheel in each image. The anomaly class was restricted to localized through-holes in the wheel skin, with variations in shape, size, and position. Broken grousers and other failure modes remained outside the scope of this work. Normal samples were defined by the absence of an injected perforation so that dust and non-anomalous surface wear could remain part of the normal visual distribution.

The annotation requirements reflected the distinction between detecting damage and localizing it. Each sample required an image-level condition label and a target-wheel mask, while anomalous samples additionally required a binary mask identifying the damage region. The training split was required to contain only normal samples, whereas validation and test included both normal and damaged conditions. For evaluation, paired clean and damaged images were generated with the same camera, wheel pose, lighting, and surface-wear parameters, allowing direct comparisons in which the only difference was the introduced perforation. Each pair had to remain inside a single split to prevent paired scene leakage. Coverage of camera viewpoints, wheel rotations, illumination, and benign wear was required to avoid evaluating inspection under a single fixed appearance.

These requirements define a controlled synthetic benchmark for visible wheel perforations, used to evaluate how efficiently the proposed approaches tackle the task.

1.2 Assets preparation

Asset preparation focused on visually representing Curiosity and recreating its Martian surroundings as faithfully as possible, while keeping the scene suitable for controlled, wheel-specific modifications. The three-dimensional scene was assembled in Blender, which was used to prepare the rover and terrain assets and render the synthetic images.

Rover model. The rover asset was based on the publicly available Curiosity model distributed by NASA’s Visualization Technology Applications and Development team [29]. Using an existing model avoided reconstructing the vehicle from photographs and provided both the wheel geometry and the surrounding rover structure. Retaining this context was important because the chassis and suspension contribute to the appearance of inspection images through occlusions and shadows.

After importing the model into Blender, the six wheels were separated into independently addressable objects, while the remaining vehicle structure was retained as an assembly. This organization enabled wheel-specific damage generation, rotation control, and mask extraction without rebuilding the rover geometry. The model was used for visual image generation rather than as an engineering representation of the rover’s mechanical behaviour.

Martian environment. The terrain was derived from HiRISE observations of Gale Crater, using a digital terrain model and associated orthorectified imagery [30]. These products provided a real observation-based foundation for the environment, including variations in elevation and surface appearance. A common region was extracted from both products, with the elevation data converted into a terrain mesh at metric scale and the corresponding imagery mapped onto its surface.

The orbital data were not detailed enough for close-up wheel views, so a local procedural terrain patch was added with grains, rocks, and small surface irregularities. Material effects provided finer visual detail without adding geometry, while rock models were reused to keep the scene manageable. These additions complemented the real terrain data but did not reproduce the exact ground surface at the site. Figure 1.1 shows the rover asset and the surrounding terrain assembled in Blender for synthetic image generation.

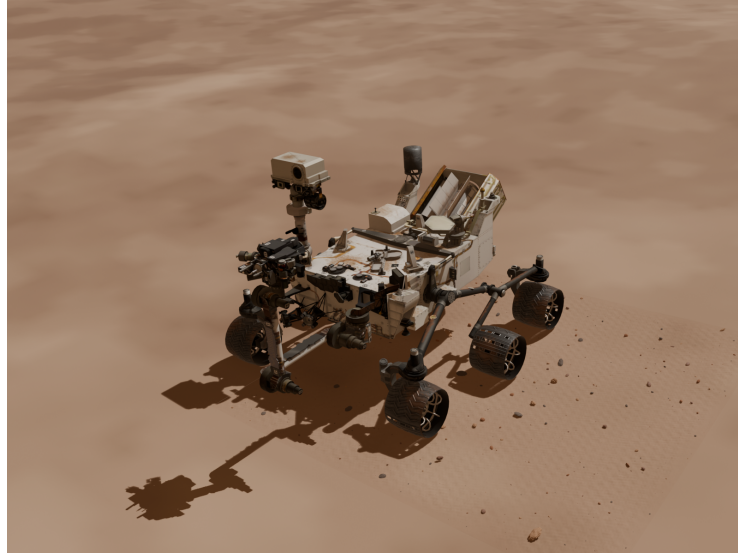


Figure 1.1. Overview of the Blender scene, showing the Curiosity rover asset and the surrounding terrain used for synthetic image generation.

1.3 Synthetic Damage Generation

Damage generation focused on a single anomaly class: through-holes in the wheel surface, with each anomalous image containing one injected hole on the designated target wheel. Hole contours were generated procedurally using three profile families: jagged slits, branched tears, and peeled-window profiles. Variations in contour irregularity, orientation, and dimensions provided different realizations within each family. Three size-based severity levels, denoted small, medium, and large, with their length sampled within 2.7-4.1 cm, 4.4-6.8 cm, and 7.3-11.6 cm, respectively. These categories describe the generated geometry rather than a mechanical assessment of wheel health. Figure 1.2 shows an example of each level.

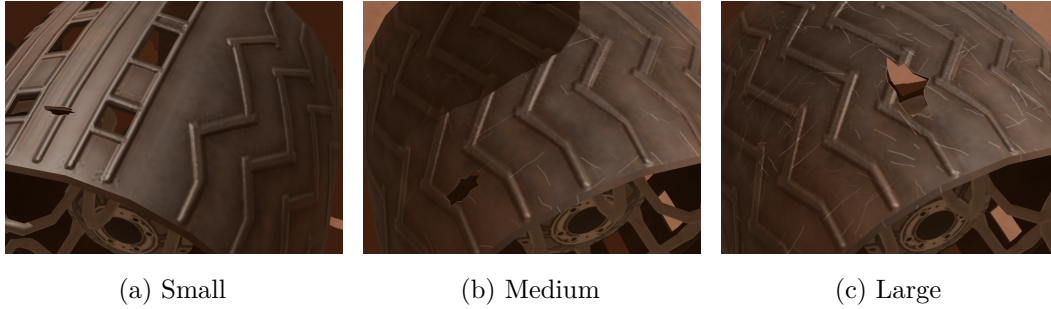


Figure 1.2. Examples of synthetic wheel damage at the three severity levels. The categories refer to the dimensions of the generated hole.

Each hole forms an opening through the wheel surface, allowing the terrain or other rover components behind it to be seen, depending on the viewpoint. Its appearance therefore also depends on the surrounding scene. The hole position varied across the tread and outer shoulder. Placement and wheel orientation were constrained so that the complete opening remained inside the image and was not hidden by the terrain or other rover components. For each damaged image, a corresponding binary mask was generated automatically, providing pixel-level annotations without manual labeling.

1.4 Domain Randomization

Domain randomization varies simulation conditions to expose models to a larger range of visual appearances [31]. In this work, variations were introduced during rendering to diversify normal wheel appearance and the context in which damage was observed, with the aim of reducing reliance on specific visual cues and helping the models distinguish actual damage from changes in appearance.

Lighting and dust appearance. Two rendering presets represented clear and dusty conditions (Figure 1.3). Alongside changes in illumination, the dusty preset introduced deposited surface dust and a mild adjustment to the camera response. Solar direction, direct and ambient illumination, and exposure were also perturbed within bounded ranges. These variations change shadows, highlights, and local con-

trast, requiring the detector to distinguish an opening in the wheel from appearance changes that can occur without damage.

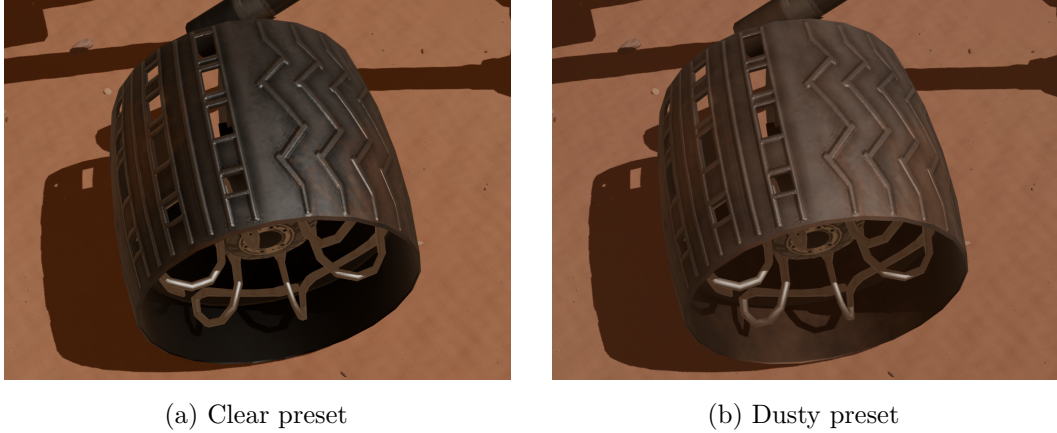


Figure 1.3. Clear and dusty rendering presets applied to the same normal wheel, with identical camera pose and wheel orientation. The dusty preset also changes deposited dust and camera response, so the comparison does not isolate illumination alone.

Normal surface variation. Three surface conditions were considered: no wear, light wear, and evident wear (Figure 1.4). Additional scratches changed the surface appearance without creating holes or altering the wheel silhouette. Surface wear did not determine the anomaly label: a sample was anomalous only when a hole was introduced, and scratches were excluded from the anomaly mask. Including these patterns among normal appearances was intended to discourage false alarms caused by high-contrast marks or local texture changes, simulating real operational wear.

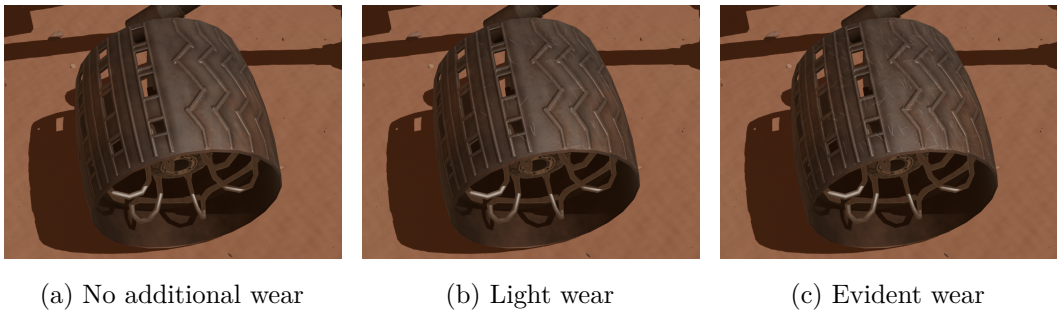


Figure 1.4. Normal surface conditions under identical lighting, camera pose, and wheel orientation. All three examples contain no injected hole and belong to the normal class.

Wheel selection and rotation. The target wheel was varied across all six rover wheels, introducing different surrounding suspension and chassis structures. For standalone normal samples, a shared rotation was applied to all six wheels using eight phases separated by 45° . Rotation changes the visible arrangement of tread features and surface marks, reducing dependence on a fixed angular configuration. For

damaged samples, wheel orientation was constrained by the visibility requirements described in Section 1.3.

Procedural ground detail. The local terrain incorporated procedurally generated relief, grains, rock fragments, and fine surface texture. These features introduce background clutter and cast shadows near the wheel, which must be distinguished from the target damage. The generated terrain realization was kept throughout dataset rendering rather than regenerated for each sample, for efficiency purposes. Its projected appearance nevertheless varied with camera pose and illumination.

Camera poses. Four wheel-centric camera configurations were used: an overhead view (A), leading and trailing three-quarter views (C1 and C2), and an upper detail view (D), as shown in Figure 1.5. The first three configurations included the complete wheel, whereas D intentionally captured a closer, partial view. Small perturbations in camera position, viewing direction, and focal length provided variation within each configuration. Changing viewpoint modifies perspective, the visible wheel surface, and the apparent size of a hole. Whole-wheel views retain more context but represent small defects with fewer pixels, while the detail view magnifies local features. The different camera configurations were also included to compare detection and localization performance across viewpoints and identify the pose best suited to wheel-damage inspection.

Within each clean/hole pair, lighting, surface wear, camera settings, and wheel orientation were kept identical. Only the injected hole distinguished the two scene states, avoiding appearance differences that could otherwise provide unintended label cues within a pair.

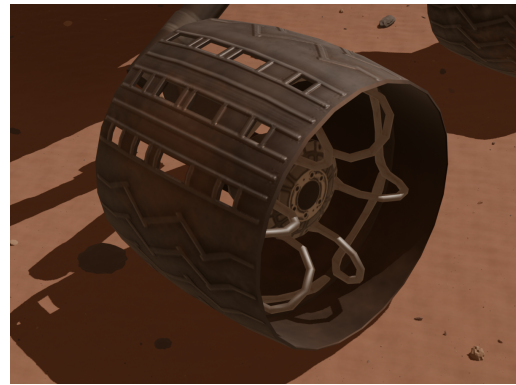
1.5 Quality Assurance

Quality assurance combined automated checks with visual inspection of selected renders. Geometric and annotation constraints determined sample acceptance, while contrast diagnostics identified potentially challenging examples without automatically excluding them. The target wheel was checked for appropriate framing and unwanted occlusions. Full-wheel views required the wheel to remain within the image, whereas the partial framing of pose D was intentional and therefore admissible. For anomalous samples, the opening and its complete boundary had to remain visible. Additional checks verified wheel-ground contact and ensured that the ground region covered by the camera remained inside the detailed terrain patch, avoiding exposed patch boundaries or unsupported wheel placement.

Damage masks were required to be spatially consistent with the target-wheel region, and free of random connected components, while severity-dependent thresholds prevented annotations from becoming excessively small. Matched normal and anomalous images were compared to assess the appearance change produced by the anomalous detail and low contrast cases were kept and flagged, provided that the remaining acceptance checks were satisfied.



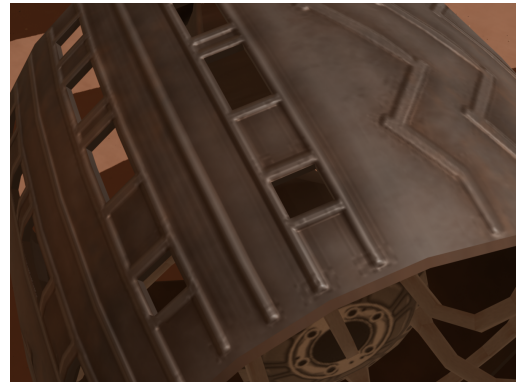
(a) A – Overhead



(b) C1 – Leading three-quarter



(c) C2 – Trailing three-quarter



(d) D – Upper detail

Figure 1.5. The four camera configurations shown for the same normal wheel under the dusty preset, without additional wear. Wheel orientation is unchanged across views.

1.6 Final Dataset Composition

The final dataset contains 10,000 RGB images, comprising 7,500 normal samples and 1,250 matched normal-anomalous pairs. This results in 8,750 normal images and 1,250 anomalous images, each containing one injected hole. Every image is associated with a target-wheel mask, while anomalous images additionally include a binary damage mask.

Dataset splits. The training set contains only normal images, allowing the anomaly-detection models to learn wheel appearance without exposure to injected defects. Validation and test sets include both conditions, as summarized in Table 1.1. Validation contains 500 standalone normal images and 250 pairs, whereas the test set consists of 1,000 pairs. Split assignments were established before rendering, and each pair was treated as an indivisible unit, meaning that an anomalous image and its normal counterpart could not be assigned to different subsets.

Split	Normal	Anomalous	Total
Training	7,000	0	7,000
Validation	750	250	1,000
Test	1,000	1,000	2,000
Total	8,750	1,250	10,000

Table 1.1. Composition of the synthetic dataset across training, validation, and test splits.

Category distributions. The main category distributions are reported in Table 1.2. The six rover wheels are represented approximately uniformly, with 1,666–1,667 images per wheel. Defect severity and placement distributions refer only to anomalous images; large defects are restricted to the tread.

Factor	Categories	Share (%)
Camera pose	A / C1 / C2 / D	40 / 27.5 / 27.5 / 5
Lighting	Dusty / clear	75 / 25
Surface wear	Current / light / evident	20 / 45 / 35
Defect severity	Small / medium / large	50 / 40 / 10
Defect location	Tread / outer shoulder	80 / 20

Table 1.2. Main category distributions in the final dataset. Camera pose, lighting, and surface wear percentages refer to all 10,000 images; defect severity and location percentages refer to the 1,250 anomalous images.

Rendering procedure. Images were generated through an automated batch pipeline in Blender 5.2 using Eevee, at a resolution of 1200×900 pixels with 64

rendering samples, and saved as 8-bit RGB PNG files. For each sample, the prepared scene was restored to its reference state before applying the parameters specified by the generation plan. Standalone normal samples required one render, whereas paired samples required a normal and an anomalous render with matching camera, illumination, surface appearance, and wheel configuration. Annotation information was obtained alongside the RGB renders, without separate mask-only renders; the normal target-wheel mask was retained for both members of each pair.

Chapter 2

Synthetic Data-Driven Damage Inspection

2.1 Task Definition

The second branch of the proposed framework addresses visual wheel inspection through the detection and localization of damage in RGB images. The experiments use the synthetic Curiosity wheel dataset introduced in the previous chapter, with localized holes as the target anomaly class. The objective is both to distinguish normal from damaged observations and to identify the image regions associated with the damage. Accordingly, the methods produce an image-level anomaly score and a spatial anomaly map, allowing detection and localization to be examined separately.

The problem was approached through visual anomaly detection using a normal-only training protocol, consistent with the setting adopted by benchmarks such as MVTec AD [33]. Only normal images from the generated dataset were used for training, while the rendered damaged images and their annotations were reserved for validation and testing. This choice was motivated by the aim of reducing dependence on annotated damage examples during training, rather than fitting a supervised segmentation model directly to the available hole masks. Normality was defined by the absence of the modeled perforations, not by a clear wheel surface, so that other appearance variations remained part of the normal distribution.

Several deep learning-based anomaly detection approaches were investigated to compare their ability to identify localized defects without confusing them with the appearance changes introduced during dataset generation. Compact configurations were also considered in view of the longer-term possibility of onboard inspection. The resulting study provides a controlled assessment of damage detection and localization within the synthetic domain, while validation on real rover imagery remains outside the scope of the present experiments.

2.2 Pre-processing

2.3 Anomaly Detection Models

2.4 Evaluation Metrics

Chapter 3

Evaluation of Lightweight Scene Understanding

3.1 Experimental Setup

3.2 Segmentation Accuracy

3.3 Quantization Results

3.4 Ablation Studies

Chapter 4

Evaluation of Synthetic Damage Inspection

4.1 Experimental Setup

4.2 Model Comparison

4.3 Ablation Studies

Chapter 5

Discussion and Limitations

5.1 Main Findings

5.2 Sym-to-Real Gap

Chapter 6

Conclusions and Future Work

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